2nd week of Development.



LEARNING OBJECTIVES

- At the end of the lecture, student should be able to

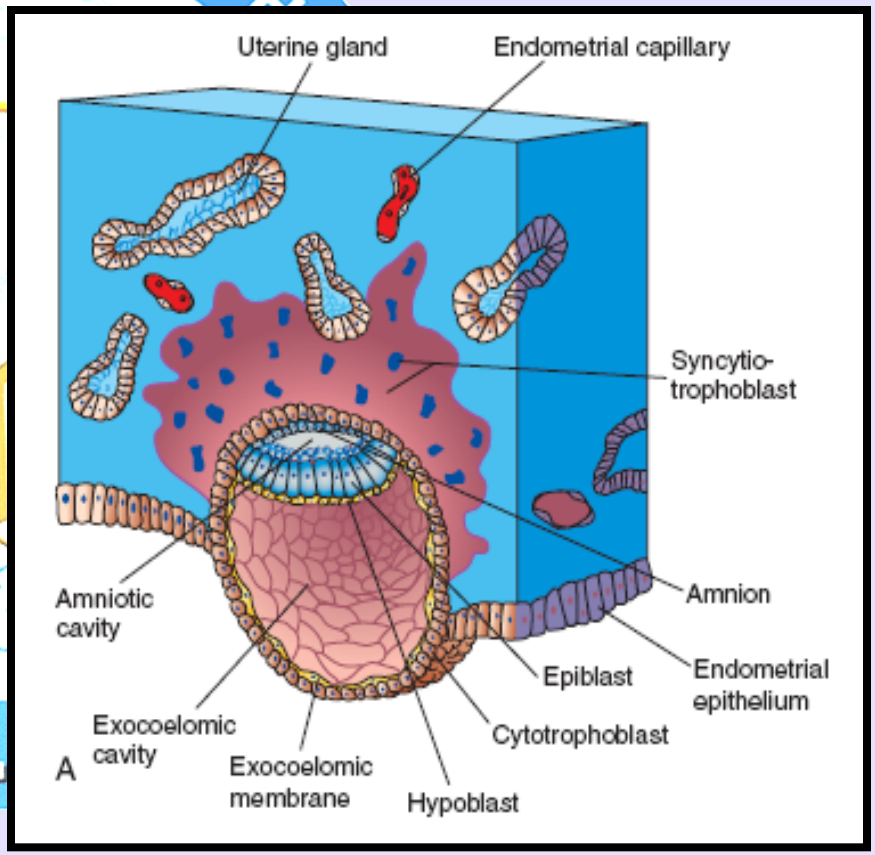
- Define implantation
- State its normal site
- Explain the formation of outer and inner cell masses
- Discuss the further development of outer cell mass (trophoblast),
- Differentiate syncytiotrophoblast and cytotrophoblast with its microscopic appearance
- Describe the process of implantation (day wise change)
- State the differentiation of embryonic pole and development of bilminar germ disc with formation Epiblast and hypoblast, their cavities (amniotic cavity and primary yolk sac)
- Discuss the development of the chorionic sac and formation Primary chorionic villi
- Enumerate the abnormal sites for implantation (ectopic pregnancy) and the different diagnostic tools

IMPLANTATION

Definition:

- It is the attachment of blastocyst to the endometrial layer of the uterine wall.

- Implantation of the blastocyst is completed during the second week.
- Morphologic changes in the embryoblast produce a bilaminar embryonic disc
- Composed of:
  - epiblast
  - hypoblast .



COMMON SITE OF IMPLANTATION

- Is posterior or anterior wall of the body of uterus
- Can be detected by
  - Ultrasonography
  - highly sensitive radioimmunoassay of hcg (as early as the end of the second week).

FORMATION OF OUTER & INNER CELL MASS

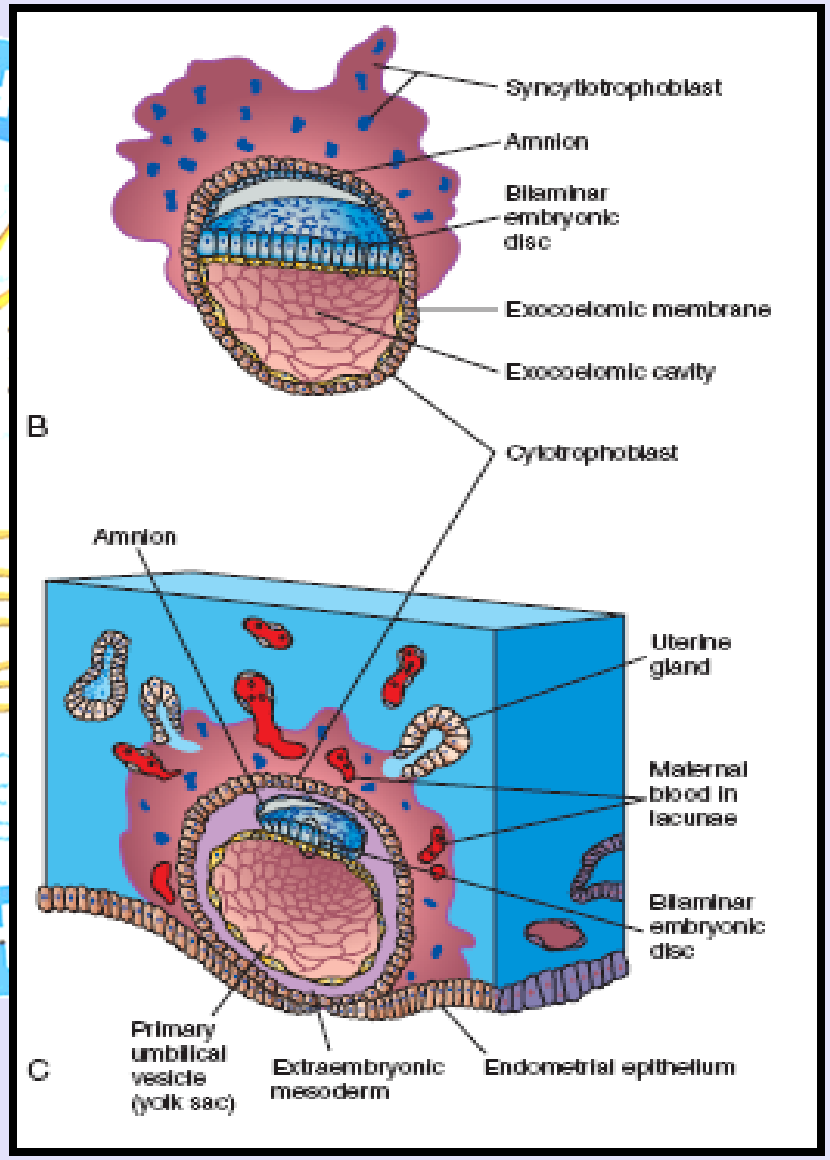
- **The inner cells mass** :Cells of the blastocyst differetiate into two layers
  1. **Layer of tall columnar cells** called epiblast adjacent to the amniotic cavity
  2. **Layer of low cuboidal** cells called hypoblast adjacent to the blastocyst cavity
- **Outer cell mass :**also differentiate into two layers
  1. **Layer of mononucleated cells** called cytotrophblast. These cells are active mitotically
  2. **Layer of multinucleated zone** without distinct cell bounderies called syncytotrophblast or syncitium

Extraembryonic structures

- The **embryonic disc** gives rise to the germ layers that form all the tissues and organs of the embryo.
- Extraembryonic structures forming during the second week are the amniotic cavity, amnion, umbilical vesicle (yolk sac), connecting stalk, and chorionic sac.

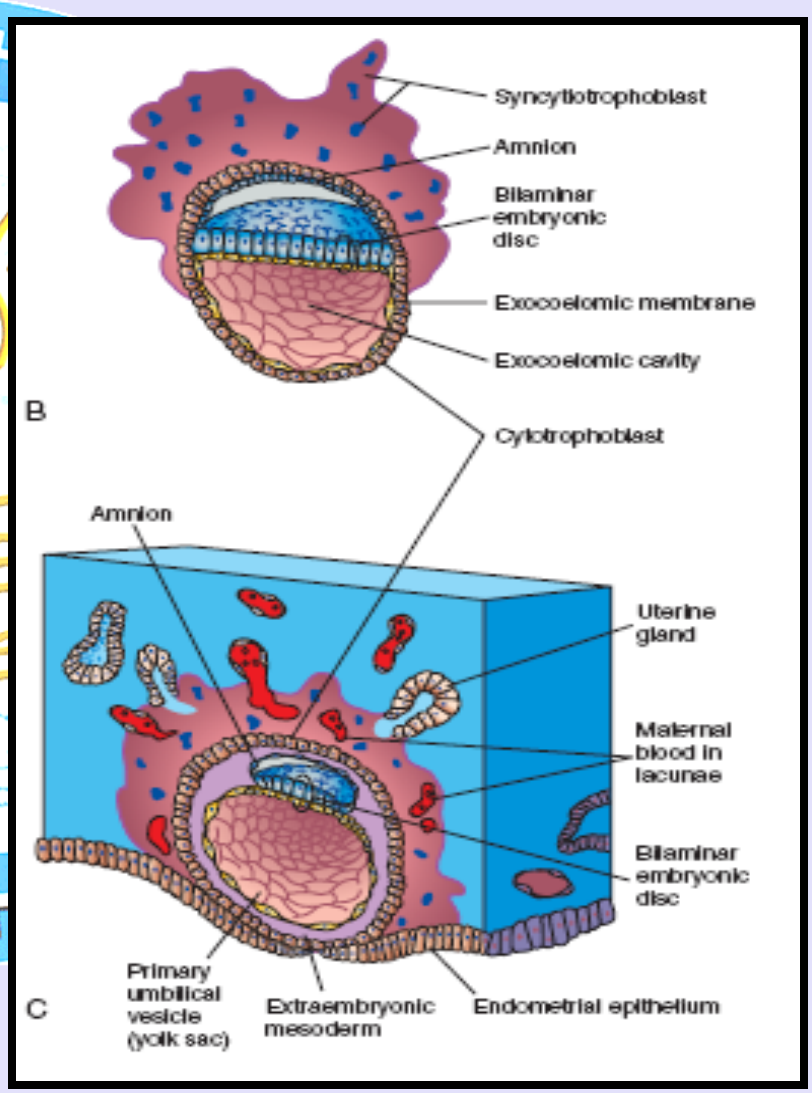
DEVELOPMENT OF TROPHOBLAST

- Implantation is completed by the end of the second week.
- It occurs during 6 to 10 days after ovulation.
- **Trophoblast differentiates into:**
  - **The cytotrophoblast:**
    - forms new cells that migrate into syncytiotrophoblast.
  - **The syncytiotrophoblast:**
    - Multinucleated mass in which no cell boundaries are discernible.



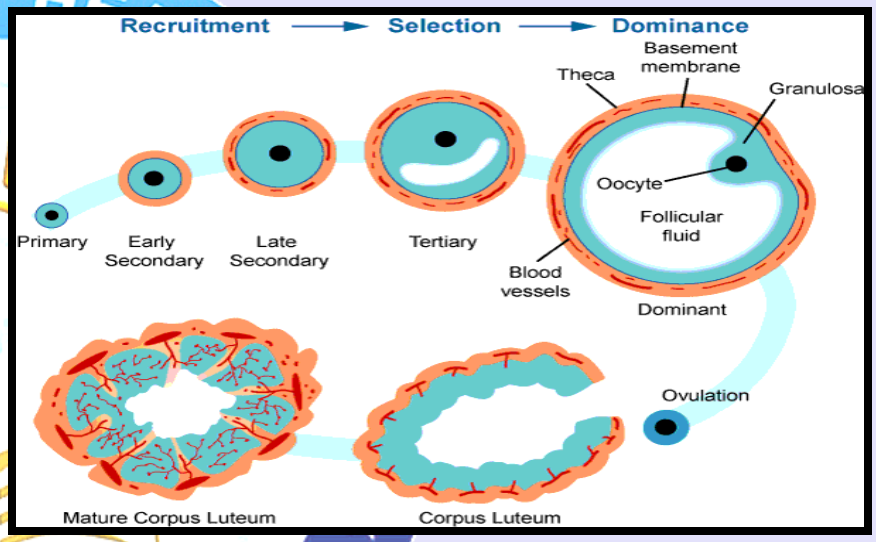
DEVELOPMENT OF TROPHOBLAST

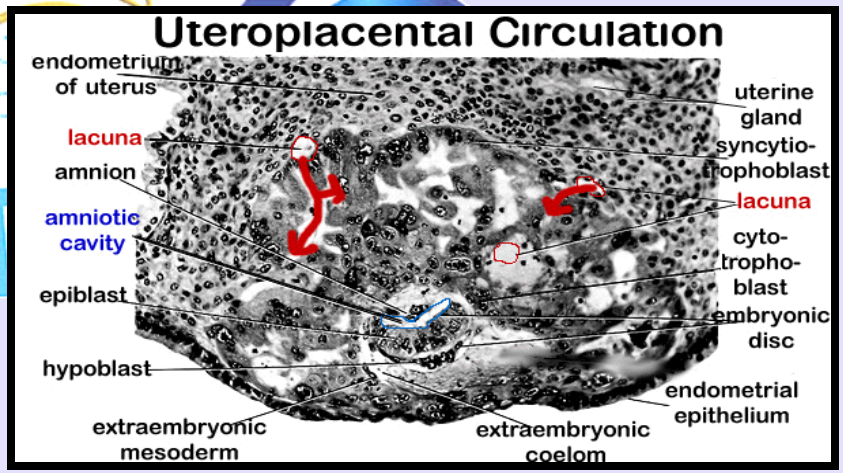
- **syncytiotrophoblast** invades the endometrial connective tissue, and the blastocyst slowly embeds itself in the endometrium.
- Syncytiotrophoblastic cells displace endometrial cells at the implantation site.
- endometrial cells undergo **apoptosis,** which facilitates the invasion.



UTERO-PLACENTAL CIRCULATION

- **VACOULES** appear within the syncytotrophoblast which collapse to form lacunae (lacunar stage)
- syncytiotrophoblast penetrate deeper into the maternal endometerium & erode the endothelial wall of the maternal cappillaries(sinusoids) & due to this maternal blood flow into the lacunar system & this starts the utro- placental circulation,
- The corpus luteum is an endocrine glandular structure that secretes estrogen and progesterone to maintain the pregnancy.

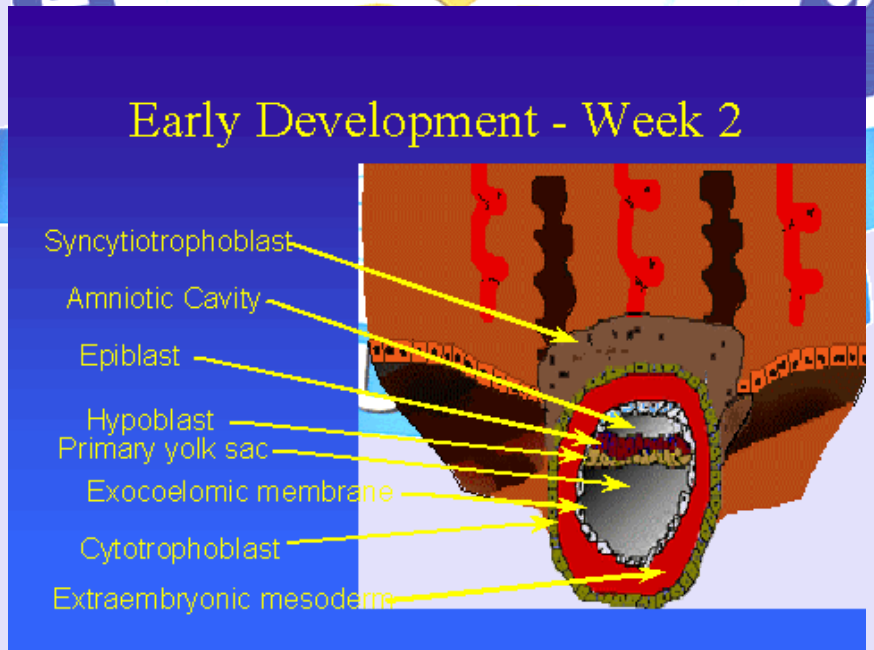




SUMMARY OF IMPLANTATION

- **Implantation of the blastocyst begins at the end of the first week and is completed by the end of the second week.**
  - The zona pellucida degenerates (*day 5*).
  - The lytic enzymes are released from the acrosomes of the sperms that surround and partially penetrate the zona pellucida.
  - The blastocyst adheres to the endometrial epithelium (*day 6*).

- The trophoblast differentiates into two layers:
  - The syncytiotrophoblast and
  - cytotrophoblast (*day 7*).
- Blood-filled lacunae appear in the syncytiotrophoblast (*day 9*).
- Lacunar networks form by fusion of adjacent lacunae (*days 10 and 11*).



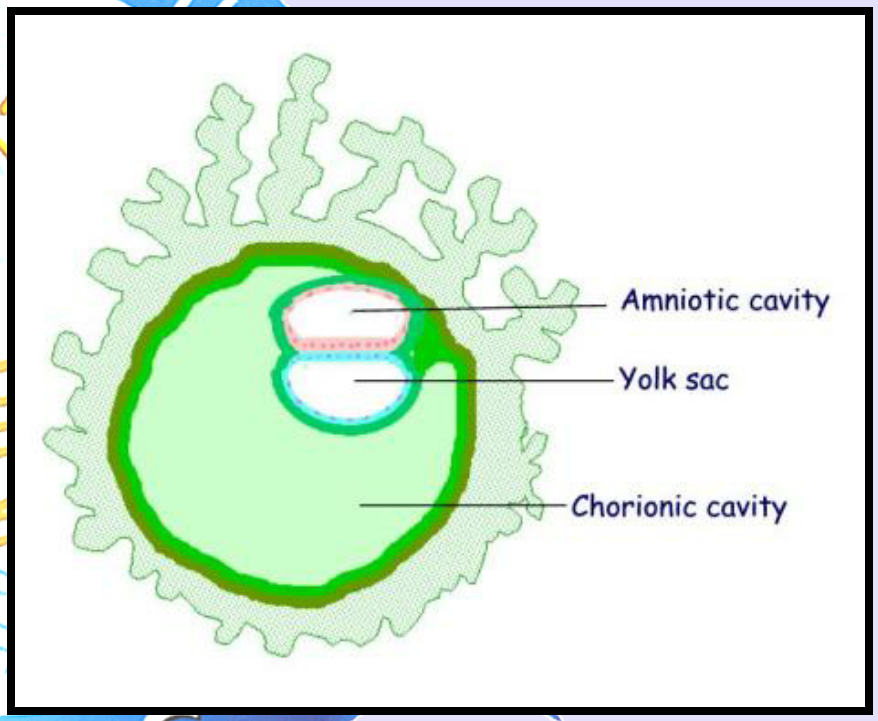
- The syncytiotrophoblast erodes endometrial blood vessels, allowing maternal blood to seep in and out of lacunar networks, thereby establishing a uteroplacental circulation (*days 11 and 12*).
- The defect in the endometrial epithelium is repaired (*days 12 and 13*).
- Primary chorionic villi develop (days 13 and 14).

NUTRITIVE SUBSTANCES

- *Oxygenated blood* passes into the lacunae from the *spiral endometrial arteries*, and *poorly oxygenated blood* is removed from them through the endometrial veins.
- *The 10-day embryo and Extraembryonic membranes is completely embedded in the endometrium* .

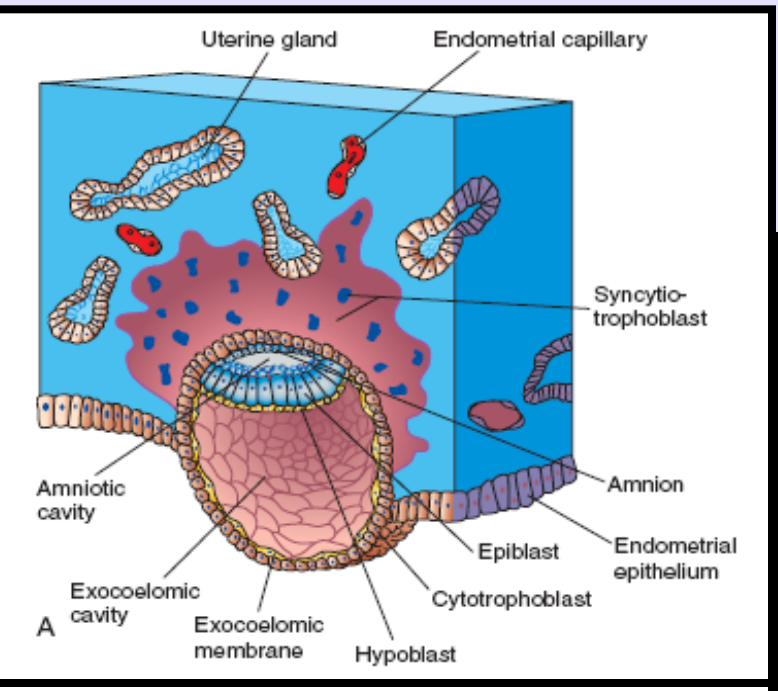
FORMATION OF THE AMNIOTIC CAVITY,

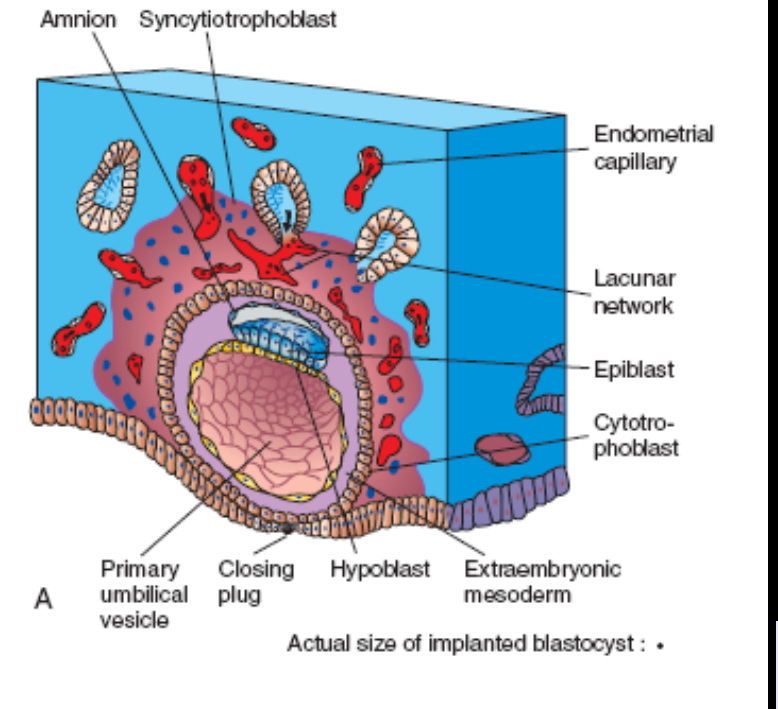
- As implantation of the blastocyst progresses, a small space " the primordium of the **amniotic cavity"** appears in the embryoblast.
- ***Amnioblasts***—separate from the epiblast and form the **amnion**, which encloses the amniotic cavity.

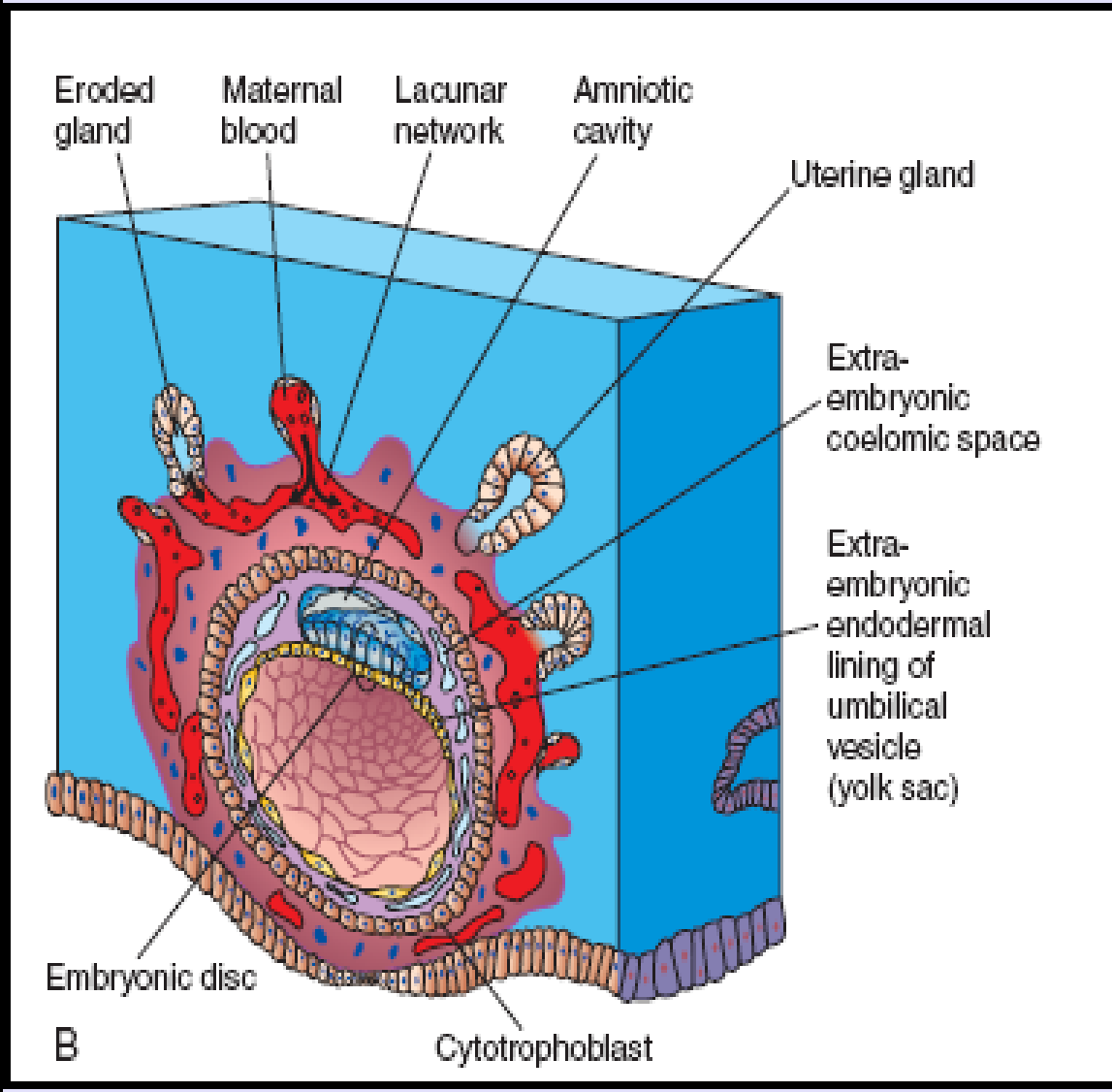


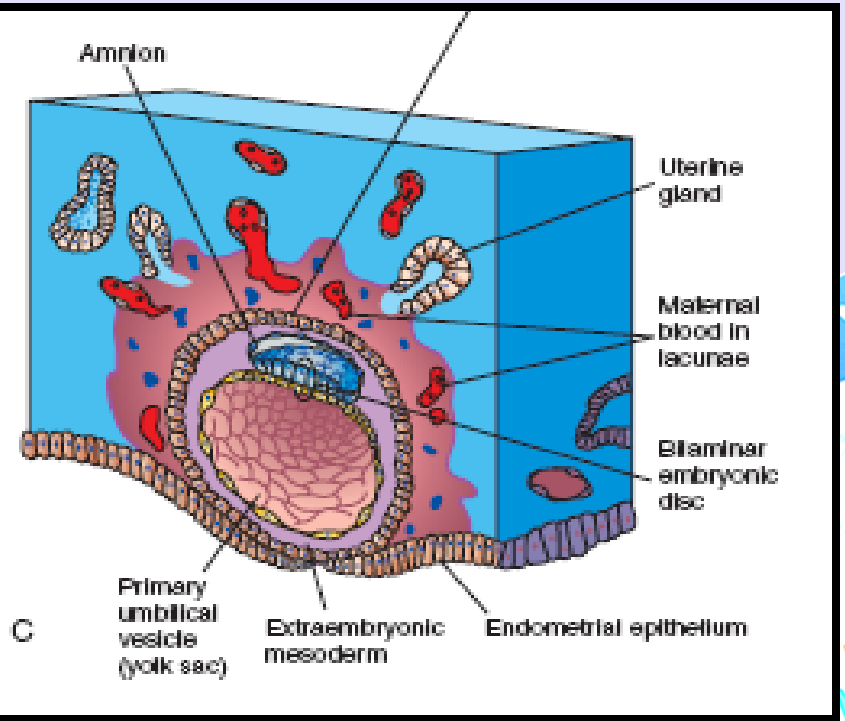
FORMATION OF EMBRYONIC DISC, AND UMBILICAL VESICLE

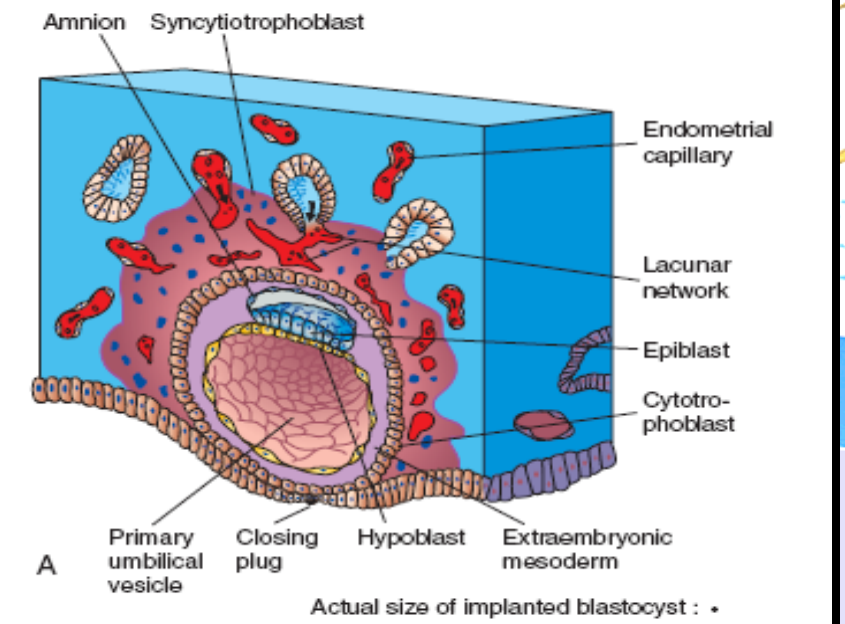
- Embryoblast results in the formation of **embryonic disc**, consisting of two layers:
  - **Epiblast :**
    - forms the floor of the amniotic cavity and is continuous peripherally with the amnion
  - **Hypoblast:**
    - forms the roof of the **exocoelomic cavity** and is continuous with the thin exocoelomic **membrane**.
    - This membrane, together with the hypoblast, lines the **primary umbilical vesicle**.


Uterine gland
Endometrial capillary
Syncytio-
trophoblast
Amniotic
cavity
Amnion
Epiblast
Endometrial
epithelium
Exocoelomic
cavity
Cytotrophoblast
A
Exocoelomic
membrane
Hypoblast


Amnion
Syncytiotrophoblast
Endometrial
capillary
Lacunar
network
Epiblast
Cytotro-
phoblast
Primary
umbilical
vesicle
Closing
plug
Hypoblast
Extraembryonic
mesoderm
A
Actual size of implanted blastocyst : •


Eroded
gland
Maternal
blood
Lacunar
network
Amniotic
cavity
Uterine gland
Extra-
embryonic
coelomic space
Extra-
embryonic
endodermal
lining of
umbilical
vesicle
(yolk sac)
Embryonic disc
B
Cytotrophoblast

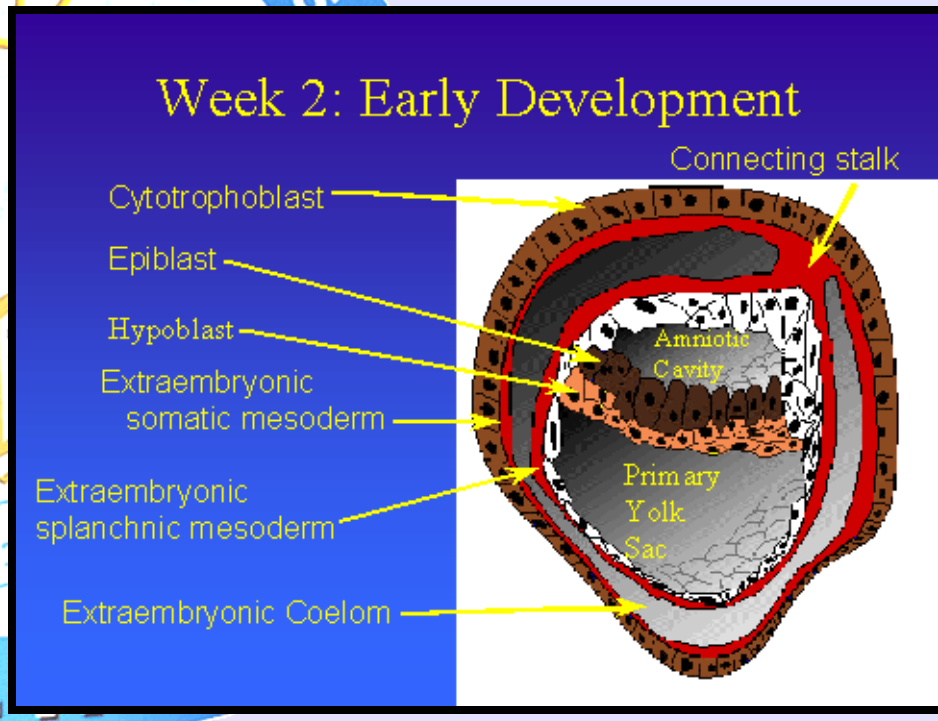


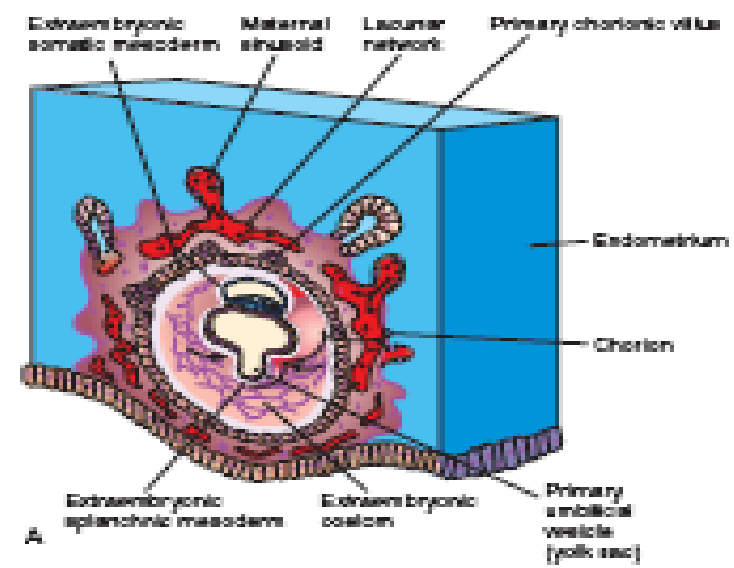


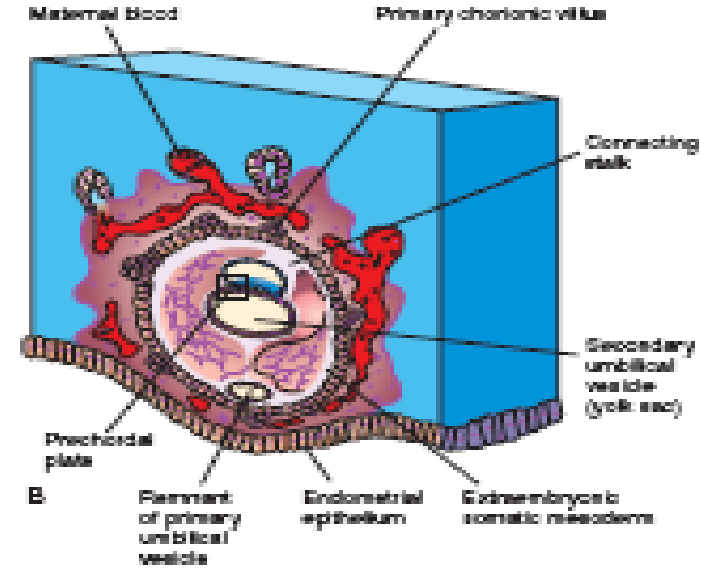
- The embryonic disc now lies between the amniotic cavity and the umbilical vesicle.
- Cells from the vesicle endoderm form a layer of connective tissue, the **extraembryonic mesoderm.**

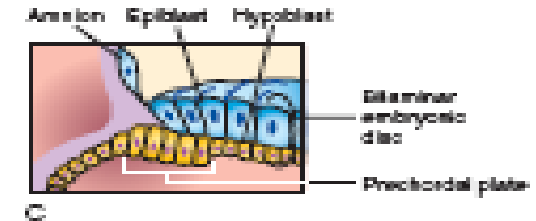
THE EXTRAEMBRYONIC COELOM

- **The extraembryonic coelom** splits the Extraembryonic mesoderm into two layers.
  - **Extraembryonic somatic mesoderm**, lining the trophoblast and covering the amnion
  - **Extraembryonic splanchnic mesoderm**, surrounding the umbilical vesicle









DEVELOPMENT OF CHORIONIC SAC

- **Primary chorionic villi appear at** end of the second week.
- Growth of cytotrophoblastic cellular extensions (finger like) is thought to be induced by the underlying **extraembryonic somatic mesoderm**.

CHORION

- The extraembryonic somatic mesoderm and the two layers of trophoblast form the **chorion** .
- The *chorion forms the wall of the chorionic sac*, within which the embryo and its amniotic sac and umbilical vesicle are suspended by the connecting stalk.
- Extraembryonic coelom is now called the **chorionic cavity**.

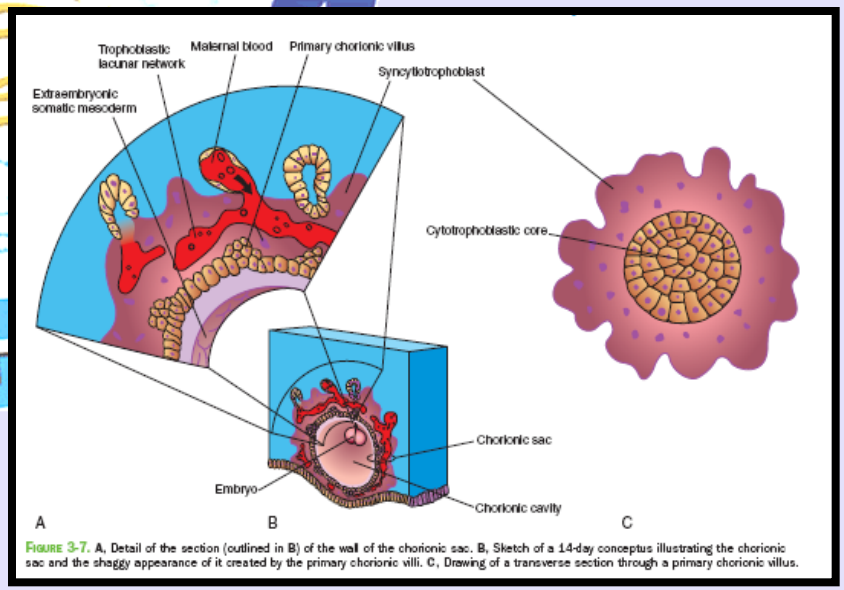


FIGURE 3-7. A, Detail of the section (outlined in B) of the wall of the chorionic sac. B, Sketch of a 14-day conceptus illustrating the chorionic sac and the shaggy appearance of it created by the primary chorionic villi. C, Drawing of a transverse section through a primary chorionic villus.

Amniotic sac

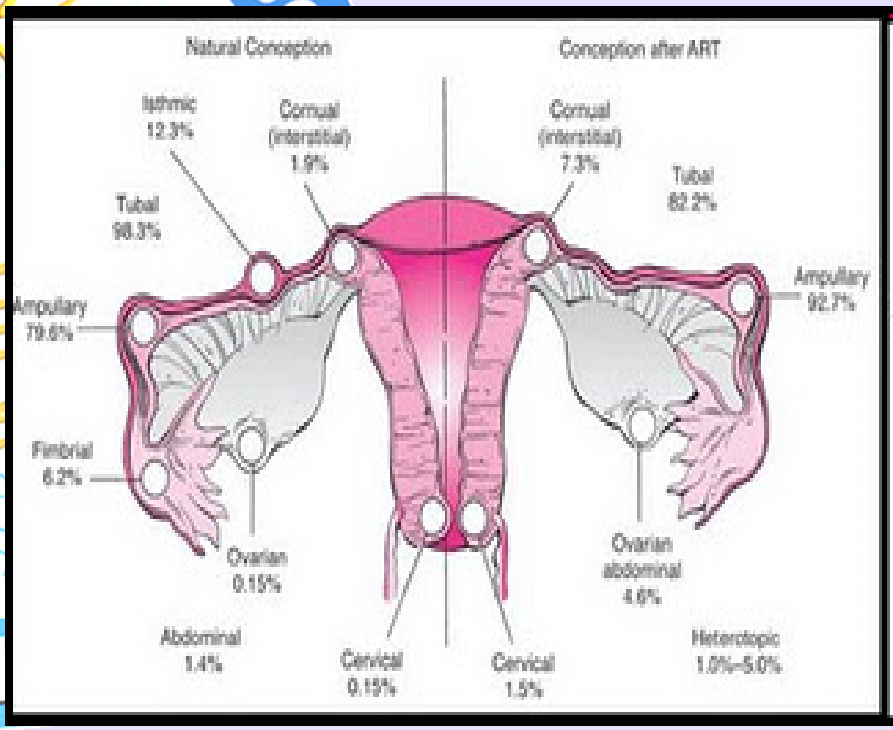
- The amniotic sac and the umbilical vesicle can be thought of as two balloons pressed together (at the site of embryonic disc) and suspended by a cord (connecting stalk) from the inside of a larger balloon (chorionic sac).

14-day embryo

- **14-day embryo** has the form of a flat bilaminar embryonic disc, but the hypoblastic cells in a localized area are now columnar and form a thickened circular area—the **prechordal plate** which indicates the future site of the mouth and an important organizer of the head region.

ABNORMAL SITES OF IMPLANTATION (ECTOPIC PREGNANCY)

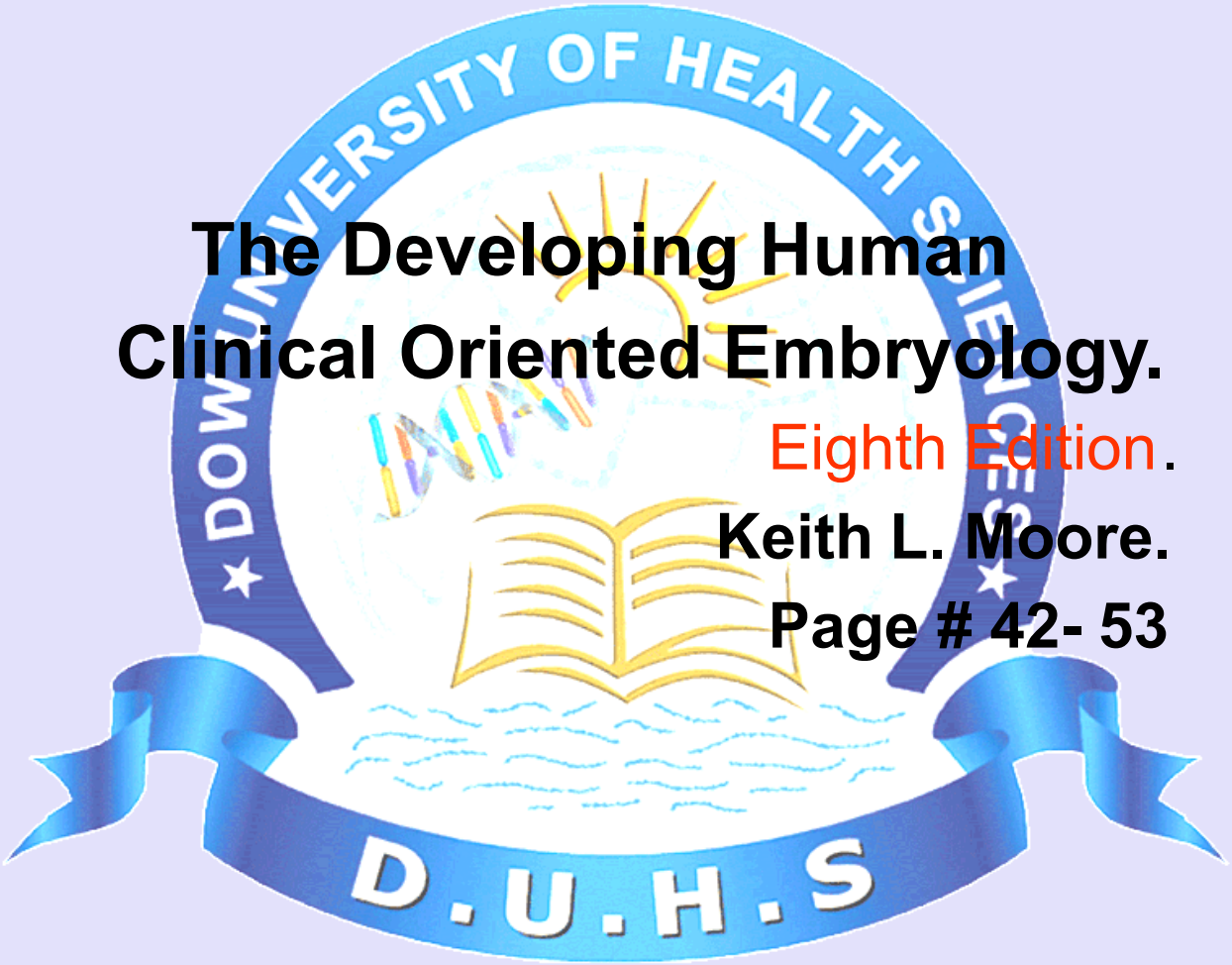
- Uterin tubes:(95% -98%)
  - » Ampulla
  - » Isthmus
- Abdominal pregnancy (commonly in rectouterine pouch)
- Cervical implantations (unusual)



DIAGNOSTIC TOOLS FOR DETECTION ECTOPIC PREGNANCY

- Transvaginal ultrasonography
- Consequent β- human chorionic gonadotropin assay

References



The Developing Human

Clinical Oriented Embryology.

Eighth Edition.

Keith L. Moore.

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THANK YOU